Multivariable Calculus @ JHU 2026

Lecture: 1

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Topics: Vector spaces, linear maps, and Euclidean space

Introduction: Linear Algebra

In this course, our goal is to develop a theory of calculus in all dimensions $n \geq 1$. To that end, we must first get to know the worlds in which we will be living. At first, this world will be the rectilinear, flat landscape of Euclidean n-space, $\mathbb{R}^n$. Later on in the course, we will seek to generalize our calculus methods to the more mysterious curved world of manifolds. But for now, let us jump first into the already rich and fascinating world of Euclid:

Section I: Vector Spaces

Definition: Euclidean n-space, denoted by $\mathbb{R}^n$ is defined as the set of all n-tuples $(x^1, x^2, \dots, x^n)$, where each "coordinate" x^i is itself just a real number. Keep caution that these are indices, not exponents! There is a good reason for this convention which will become clear later.

We typically call such an n-tuple a point, and use the shorthand notation

$$x = (x^1, x^2, \dots, x^n).$$

It is a key fact that $\mathbb{R}^n$ has an additional algebraic structure, in that $\mathbb{R}^n$ can be seen as a vector space:

Definition: A vector space is a quintuple of data $(V, \vec{0}, \mathbb{F}, +, \cdot)$ where

- V is a set of elements, called vectors;
- $\mathbb{F}$ is a field (here we always take $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) whose elements are called scalars;
- $+ : V \times V \to V$ is a binary operation on vectors, called vector addition;
- $\cdot : \mathbb{F} \times V \to V$ is a binary operation on a scalar and vector pair, called scalar multiplication;

such that the following vector space axioms are satisfied for every $x, y, z \in V$ and $\alpha, \beta \in \mathbb{F}$:

1) Commutativity of $+$: $x + y = y + x$
2) Associativity of $+$: $x + (y + z) = (x + y) + z$
3) Additive Identity: There is a unique vector $\vec{0} \in V$ such that $x + \vec{0} = x$
4) Additive Inverses: $x + (-1) \cdot x = \vec{0}$. We write $(-1) \cdot x$ as $-x$.
5) Multiplicative Identity: $1 \cdot x = x$
6) Multiplicative Associativity: $\alpha \cdot (\beta \cdot x) = (\alpha\beta) \cdot x$
7) Distributive Properties:
   - $(\alpha + \beta) \cdot x = \alpha x + \beta x$
   - $\alpha \cdot (x + y) = \alpha x + \alpha y$

Key Example (Continued from above)

In the language above, $\mathbb{R}^n$ can be seen as a real vector space (i.e. $\mathbb{F} = \mathbb{R}$) where:

- $V = \mathbb{R}^n = \{ x = (x^1, \ldots, x^n) : x^i \in \mathbb{R} \}$
- $x + y = (x^1, \ldots, x^n) + (y^1, \ldots, y^n) = (x^1 + y^1, \ldots, x^n + y^n)$
- $\alpha x = \alpha \cdot (x^1, \ldots, x^n) = (\alpha x^1, \ldots, \alpha x^n)$

Some Key General Concepts from Linear Algebra

To better understand $\mathbb{R}^n$ as a vector space (and how it serves as the right starting point for developing high dimensional calculus), it will be very useful to first introduce/recall some of the foundational results of linear algebra in the large.

Let V be a vector space as in the definition above.

Definition: A subset $W \subseteq V$ is called a linear/vector subspace of V if W satisfies the vector space axioms in its own right. It is sufficient to check that $\forall v, w \in W$ and $\mu, \nu \in \mathbb{F}$, $\lambda v + \mu w \in W$.

Definition: Let $v_1, v_2, \dots, v_k \in V$, and $\mu^1, \mu^2, \dots, \mu^k \in \mathbb{F}$. The vector $v := \mu^1 v_1 + \cdots + \mu^k v_k$ in V is called a linear combination of the v_i with scalars μ^i.

Definition: A set of vectors $\{v_1, \dots, v_k\} \subseteq V$ is said to span V if every $v \in V$ is a linear combination of the $\{v_1, \dots, v_k\}$.

Definition: A set of vectors $\{v_1, \dots, v_k\} \subseteq V$ is said to independent if no $v_i \in \{v_1, \dots, v_k\}$ is a linear combination of the others. It is equivalent to check that the only $\mu_1, \dots, \mu_k \in \mathbb{F}$ which solve the equation $0 = \mu^1 v_1 + \cdots + \mu^k v_k$ are $\mu^1 = \cdots = \mu^k = 0$. Otherwise, the set $\{v_1, \dots, v_k\}$ is said to be dependent.

Definition: A set of vectors $\{v_1, \dots, v_n\} \subseteq V$ which is both spanning and independent is called a basis of V.

In particular: If $v \in V$, then there exists a unique way to write v as a linear combination of vectors in the basis $\{v_1, \dots, v_n\}$.

Dimension Theorem:

1. Every vector space $(V, \mathbb{F})$ has a basis.
2. Any two bases of $(V, \mathbb{F})$ have the same cardinality.

Definition: This common cardinality is called the dimension of $(V, \mathbb{F})$

Convention: The trivial vector space $V = \{0\}$ has dimension 0.

In this course, we will focus on finite dimensional vector spaces over $\mathbb{R}$ (or maybe $\mathbb{C}$). The subject of infinite dimensional vector spaces in the context of calculus is the purview of functional analysis.

As we will be able to show momentarily, f.d. (finite dimension) v-spaces over $\mathbb{F}$ are of a very particular, clean form:

Theorem: Every vector space $(V, \mathbb{F})$ of finite dimension $n \in \mathbb{Z}^{\geq 0}$ is of the form $V = \mathbb{F}^n$ (ie. n-tuples $(v^1, \dots, v^n)$ with componentwise addition / scalar multiplication).

So, without any loss of generality (and with a great gain in clarity) we really only need to think about the vector spaces $\mathbb{F}^n$ in this course — in particular $\mathbb{R}^n$ and $\mathbb{C}^n$.

To better understand why this is possible, we need to continue our recollection of key linear algebra definitions / results:

Section II: Linear Transformations

Given that our spaces of interest (vector spaces) have a special algebraic structure, it is natural to study maps between these spaces which are compatible with that algebraic structure.

Definition: Let V, W be $\mathbb{F}$-vector spaces. A function $T: V \to W$ is called a linear transformation/map iff $\forall v, w \in V$, $\mu, \nu \in \mathbb{F}$,

1. $T(v+w) = T(v) + T(w)$
2. $T(\mu v) = \mu T(v)$

In other words, T takes linear combinations to linear combinations:

$$T(\underbrace{\mu^1 v_1 + \cdots + \mu^k v_k}_{\in V}) = \underbrace{\mu^1 T(v_1) + \cdots + \mu^k T(v_k)}_{\in W}.$$

Facts: Let V, W be $\mathbb{F}$-vector spaces, and $T: V \to W$ a linear map.

1. $\operatorname{im} T := \{T(v) : v \in V\} \subseteq W$ is a subspace of W
   $=:$ image of T
2. $\ker T := \{v \in V : T(v) = 0\}$ is a subspace of V.
   $=:$ kernel of T
3. **Rank-Nullity Theorem:**
   $$\underbrace{\dim(\operatorname{im}(T))}_{\text{rank of } T} + \underbrace{\dim(\ker(T))}_{\text{nullity of } T} = \dim(V).$$
4. **Basis Theorem:** Let V have basis $\{v_1, \ldots, v_n\}$ and let $\{w_1, \ldots, w_n\}$ be any n vectors in W. Then $\exists!$ linear map $T: V \to W$ such that $T(v_i) = w_i$ for all i.

Important consequence of (4): Once you know what $T: V \to V$ does to a basis $\{v_1, \dots, v_n\}$ of V, you know what T does to every vector in V: writing (uniquely!) $v = a^1 v_1 + \cdots + a^k v_n$,

$$T(v) = T(a^1 v_1 + \cdots + a^k v_n) = a^1 T(v_1) + \cdots + a^k T(v_n) = a^1 w_1 + \cdots + a^k w_n.$$

In this course, our main interest in linear mappings is due to the construction of the differential Df of a function f, which generalizes the derivative $f'(x)$ of a single variable function. Eventually, to develop the chain rule, we'll need to consider compositions of linear maps:

Facts: Let U, V, W be $\mathbb{F}$-vector spaces, with linear maps $T: U \to V$ and $S: V \to W$.

(1) $S \circ T: U \to W$ is also a linear map

(2) Let $U = W$. If $S \circ T: U \to U$ has $S \circ T = Id_U$ (the identity map $I(u) = u$ of U) we say that T has S as a left inverse, and S has T as a right inverse. If a linear map $T: U \to V$ has both a left and right inverse itself (say $S: V \to U$ so that $S \circ T = Id_U$, $T \circ S = Id_V$) then T is said to be invertible, or a linear isomorphism. We write $S = T^{-1}$. (check that this is well defined!)
If V, W are $\mathbb{F}$-vector spaces with a linear isomorphism between them, V and W are said to be isomorphic. Isomorphic vector spaces are completely equivalent from the point of view of linear algebra.

(3) If $T: U \to V$ is a linear map which is both injective and surjective, then T is a linear isomorphism.

Section III: Euclidean Space as a Vector Space

Recall that throughout, $\mathbb{F}$ means either $\mathbb{R}$ or $\mathbb{C}$.

Theorem: Every vector space $(V, \mathbb{F})$ of finite dimension $n \in \mathbb{Z}^{>0}$ is of the form $V = \mathbb{F}^n$ (ie. n-tuples $(v^1, \dots, v^n)$ with componentwise addition / scalar multiplication

Proof: Clearly $\mathbb{F}^n$ is such a vector space, as we have noted earlier. Now and forever, we'll think of the elements $(x^1, \dots, x^n) \in \mathbb{F}^n$ as column vectors instead of as row vectors:

$$v \in \mathbb{F}^n \longleftrightarrow v = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix}$$

This notational choice / trick will be very handy in everything to come, and the following argument will begin to illustrate why.

Notational note: Logistically, it's still easier to write vectors $\begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} \in \mathbb{F}^n$ as n-tuples $(x^1, \dots, x^n)$ as we have done before. The upper indices are there to remind us that this n-tuple denotes the coordinates of a vector.

Now, suppose V is a f.d. $\mathbb{F}$-vector space. We can thus find a finite basis $\{v_1, \dots, v_n\}$ of V by definition (in general, it's a deep fact that every vector space, f.d. or not, has a basis — at least if you are willing to invoke the Axiom of Choice in the proof). Therefore, every vector $v \in V$ has a unique "recipe" using the ingredients $\{v_1, \dots, v_n\}$:

$$v = v^1 v_1 + \dots + v^n v_n .$$

That is: every $v \in V$ produces a unique "identifier code" $(v^1, \dots, v^n) \in \mathbb{F}^n$, and every list of scalars $(v^1, \dots, v^n) \in \mathbb{F}^n$ produces a unique vector $v = v^1 v_1 + \dots + v^n v_n \in V$.

This defines a bijective map $\Phi: V \to \mathbb{F}^n$ by setting

$$\Phi(v) = (\mu^1, \mu^2, \dots, \mu^n).$$

We can directly check that Φ is linear: if $v, w \in V$, $\alpha, \beta \in \mathbb{F}$, suppose $\Phi(v) = (\mu^1, \dots, \mu^n)$, $\Phi(w) = (\lambda^1, \dots, \lambda^n)$. Then

$$\begin{aligned} \alpha v + \beta w &= \alpha(\mu^1 v_1 + \cdots + \mu^n v_n) + \beta(\lambda^1 v_1 + \cdots + \lambda^n v_n) \\ &= (\alpha\mu^1 + \beta\lambda^1) v_1 + \cdots + (\alpha\mu^n + \beta\lambda^n) v_n \end{aligned}$$

} the unique recipe for $\alpha v + \beta w$

so that by definition of Φ, and the v-space rules in $\mathbb{F}^n$,

$$\begin{aligned} \Phi(\alpha v + \beta w) &= (\alpha\mu^1 + \beta\lambda^1, \dots, \alpha\mu^n + \beta\lambda^n) \\ &= \alpha(\mu^1, \dots, \mu^n) + \beta(\lambda^1, \dots, \lambda^n) \\ &= \alpha\Phi(v) + \beta\Phi(w). \end{aligned}$$

Thus, $\Phi: V \to \mathbb{F}^n$ is a linear isomorphism, and we call it the **coordinate map with respect to the basis** $\{v_1, \dots, v_n\}$.



Consequently we really only need to think about the vector spaces $\mathbb{F}^n$ in this course — in particular $\mathbb{R}^n$ and $\mathbb{C}^n$. This makes everything very concrete. On the other hand, this **Theorem** tells us that we are also free to use **all of the powerful theory of linear algebra** in general to help solve problems!

Let's return, then, to our starting example of $\mathbb{R}^n$, where points are viewed as column vectors: $v \in \mathbb{R}^n \leftrightsquigarrow v = (x^1, \dots, x^n)$. Here we get concrete examples of the many definitions and concepts that we met above...

Facts about $\mathbb{R}^n$: Prove these!

(1) $\mathbb{R}^n$ has dimension n. The standard basis is defined by

$$e_1 = (1, 0, \dots, 0),\ e_2 = (0, 1, 0, \dots, 0),\ \dots,\ e_n = (0, \dots, 0, 1).$$

(2) The subspaces of $\mathbb{R}^n$ are isomorphic to $\mathbb{R}^k$ for $0 \le k \le n$.

(3) Every linear transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ is given by a matrix:

On the one hand, if

$$A = \begin{bmatrix} a_1^1 & \cdots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^m & \cdots & a_n^m \end{bmatrix}$$

is an $(m \times n)$ – $\mathbb{R}$ matrix, and $v = (x^1, \dots, x^n) \in \mathbb{R}^n$, then matrix multiplication

$$v \longmapsto A \cdot v = \begin{bmatrix} a_1^1 & \cdots & a_n^1 \\ \vdots & \ddots & \vdots \\ a_1^m & \cdots & a_n^m \end{bmatrix} \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} = \begin{pmatrix} x^1 a_1^1 + x^2 a_2^1 + \cdots + x^n a_n^1 \\ \vdots \\ x^1 a_1^m + x^2 a_2^m + \cdots + x^n a_n^m \end{pmatrix}$$

defines a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

On the other hand, if $T: \mathbb{R}^n \to \mathbb{R}^m$ is any linear transformation, pick a basis of $\mathbb{R}^n$ — say $\{e_1, e_2, \dots, e_n\}$. Let $a_i := T(e_i) \in \mathbb{R}^m$, and set

$$A := \begin{bmatrix} \vdots & \vdots & & \vdots \\ a_1 & a_2 & \cdots & a_n \\ \vdots & \vdots & & \vdots \end{bmatrix}$$

to be the $(m \times n)$ $\mathbb{R}$-matrix with the $a_i = T(e_i)$ as its columns. Then we see that $A \cdot e_i = a_i = T(e_i)$ for all $i = 1, \dots, n$, hence $T(-) = A \cdot (-)$ by the earlier Basis Theorem. Thus, every linear map $T: \mathbb{R}^n \to \mathbb{R}^m$ is just the action of a matrix!

(4) If $\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m \xrightarrow{S} \mathbb{R}^k$ are linear maps (i.e. matrices), then the matrix corresponding to $S \circ T: \mathbb{R}^n \to \mathbb{R}^k$ is the matrix product of the matrices for S and T.

(5) If $T: \mathbb{R}^n \to \mathbb{R}^n$ is an isomorphism, its matrix A is invertible, and T^{-1} has matrix A^{-1}.

Important Convention: As we have remarked above, when writing a point $x \in \mathbb{R}^n$ in coordinates, we will often write $x = (x^1, \ldots, x^n)$ (note the round brackets and upper indices) while understanding that we think of this point as a column vector

$$x = \begin{bmatrix} x^1 \\ \vdots \\ x^n \end{bmatrix}$$ (note the square brackets).

In general, when square brackets are used, we interpret them literally in the shape they are found. eg., $[x^1, \ldots, x^n]$ denotes a $(1 \times n)$ row vector, and $([x^1, x^2], [y^1, y^2])$ denotes the matrix $\begin{bmatrix} x^1 & x^2 \\ y^1 & y^2 \end{bmatrix}$ formed by putting the rows $[x^1, x^2]$, $[y^1, y^2]$ into a column.

This will be important to keep straight when we start forming involved compositions of various maps. My tip: always start interpretting an expression by type-checking. That is, consider the sort of objects involved in an expression to make sure they are compatible. eg: if $T: \mathbb{R}^n \to \mathbb{R}^m$, $S: \mathbb{R}^p \to \mathbb{R}^q$, $R: \mathbb{R}^s \to \mathbb{R}^t$, are linear transformations, and $\bar{T} \in M^{m \times n}(\mathbb{R})$, $\bar{S} \in M^{q \times p}(\mathbb{R})$, $\bar{R} \in M^{t \times s}(\mathbb{R})$ are their matrices in the standard bases,

1. $\bar{T} \cdot \bar{S} + \bar{R}$ is not defined unless $p = s$ (domains agree), $q = n$ (composition of $T \circ S$ is defined), and $t = m$ (codomains agree)
2. $R = (T, S)$ is not well formed unless $n = p = s$ (domains agree), $t = m + q$ (codomains agree). $(T,S)(v) := (T(v), S(v))$

Exercises for Lecture I:

1. Verify all the claims/exercises in the notes.
2. Check v-space axioms for some example
3. Matrix algebra refresher
4. Linear maps vs matrices — coordinate representations of maps, vectors,
5. Determinants / inverses / rank